

Medium Access Control Protocols

Module 08

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science Punjabi University patella today's lecture is on Mac protocols in this lecture we are going to discuss in detail about the working and Protocols of medium access sublayer of OSI model now let's start the medium access sublayer is the bottom part of data link layer the medium access sub layer is also known as Mac that is Media Access Control sublay when a common medium is shared by many stations Mac lab plays a very important role without this control several stations transmitting simultaneously could produce gobbled messages the media Access Control mechanism standardized by i e are implemented in the max sublayer of the data link layer it provides service to the LLC logical link control sublayer and receives service from the physical layer below it the basic

functions of Max sublayer is the media Access Control error detection and station addressing media Access Control procedures are implemented to ensure that every station gets a fair chance to transmit and collisions do not take place there are several methods of media Access Control in lands and each method is applicable to specific land topology in addition to the basic access procedures the Mac layer also handles the frame D limiting address recognition and error checking functions when a number of user stations share a single transmission medium this is called a multiple access communication the transmission medium is broadcast in nature so all other attached stations to the medium can receive the transmission From Any Given station the physical transmission medium

of a land is shared by the stations connected on the land let's see how media is 2 minutes, 1 second shared the two basic approaches to medium Access Control are static and dynamic access control or in other words it is called Static Channel allocation and dynamic Channel allocation the dynamic access control is further categorized into random access and scheduling the figure here shows various techniques for sharing a transmission medium first we'll talk about static Channel allocation in land number of computers are working and only one computer is allowed to broadcast on the communication channel the first method of allocating a single Communication channel among multiple contending users is by using frequency division multiplexing that is

fdm if there are n number of users the entire bandwidth is divided into n equal partitions and each user is allocated one partition since each user has been allocated a fixed frequency band 3 minutes, 1 second therefore interference between them is avoided but this scheme is applicable only when number of users are less and each has constant traffic when number of users in the land is large with constantly waiting traffic then the bandwidth is divided equally into n partitions if the number of users is less than n then a significant portion of frequency spectrum is wasted but if the number of users is greater than n then some of them will not get a frequency Channel due to shortage of bandwidth in this case channel is set to be inefficient if all the users who want

to communicate are arranged in one q and number of users are n then the average delay T will be n times the meantime delay the static fdm exhibits poor performance of Channel next method of Channel allocation is dynamic Channel allocation Dynamic Channel allocation method is included in all lands and vans Dynamic Channel allocation method is able to handle all the different types of traffic conditions an order is built up among the Stations of the land so that proper chance is given to each station to transmit its data frames but for this some requirements are to be satisfied by the land first requirement is that the workstations or terminals are independent and work is generated at a constant rate second requirement is that a single channel is available for all communication all stations can transmit

and receive through it third requirement is that the workstation can begin signal transmission at any time or in a slotted time assigned to it fourth requirement is that if two stations are transmitting simultaneously the signals collide with each other and resulting signal is gobbled this process is called Collision this Collision can be detected by all the other stations the collided signal may be retransmitted the last requirement is that stations can have either carrier sensing or not carrier sensing is a specific electrical signal on the network sensing of carrier gives the status of the channel That is busy or free next we are going to discuss multiple access protocols one feature of Lan is that its backbone is a shared Channel or transmission link which provides all

users to access the transmission facilities it may happen that two or more stations transmitting data at the same time causing their signals to interfere and data gets gobbled so in order to resolve these type of conflicts a number of different medium access protocols have been developed in order to handle the bursty nature of land burstiness means the data is transmitted irregularly in bursts rather than as continuous streams traffic asynchronous stedium is used the asynchronous stedum mechanism is further divided into two methods contention method or Random Access method method and deterministic Method or they are also called as controlled methods Random Access techniques are Aloha carrier sense multiple xes or

CSMA then CSMA with Collision detection which is also called CSMA/CD and the last one is register insertion it is not very popular and is obsolete. Controlled access to LAN can be performed in two types: centralized or distributed. In centralized technique, Master node decides which node is to access the channel at any point of time for example polling. In distributed technique, each station is given an opportunity to transmit on the channel for example token passing method and slotted ring method. This figure illustrates the typical multiple Access Communications where a number of user stations share a transmission medium. These sharing techniques are used both in wired and wireless communication networks in wired communication.

Multi-drop cables are used in data networks to connect number of stations to a host computer. The host computer broadcasts information to the users on the outbound line. The stations transmit information to the host using the inbound line. This system is shown in this figure. A medium access control or MAC protocol is developed for the system. Here the host computer issues polling messages to each station providing it with permission to transmit on the inbound line. In radio communication, several stations share two frequency bands: one for transmitting and one for receiving. In Satellite Communications, each station is assigned a channel in an Uplink frequency band that it uses to transmit to the satellite. The satellite sends

back the signals on different frequency band called down link frequency band. First technique is Aloha. The Aloha protocol is a contention protocol which was developed at the University of Hawaii in early 1970s and was originally developed for packet Radio Networks. However, it is applicable to any shared transmission medium in a system when many users try to send messages to each other using a common broadcast channel. Random Access or contention techniques are used. Random Access means there is no exact time for any station to transmit data. This scheme is very simple and is asynchronous. It is asynchronous because there is no proper coordination among the transmitting stations. The basic idea of Aloha system is useful to any system in which coordinated stations are competing for

the use of a single shared Channel. When a station sends data, another station may try to do so at the same time. The data from the two stations may collide and get scrambled. If there is a collision, then each station would simply wait for a random period of time before trying again. The Aloha system is of two types: Pure Aloha which does not require Global time synchronization and slotted Aloha which requires time synchronization. We will first discuss Pure Aloha. The original Aloha protocol is called Pure Aloha. In this, the whole idea is that each station can send a frame whenever it has a frame to send as there is only one channel available to share. There is the possibility of collision between frames from different stations. The Pure Aloha protocol depends upon acknowledgement from the receiver. When a

user sends a frame, it expects the receiver to send an acknowledgement. If the acknowledgement does not reach the transmitter after the timeout period, the station assumes that the frame has been destroyed and resends the frame. The figure shows the frame collisions.

in Pure Aloha as you can see when two frames are transmitted at the same time there will be a collision and both will be gobbled if the first bit of a new frame overlaps with just the last bit of a frame almost finished then both the frames will be completely destroyed and both will have to be retransmitted again after the time of period if all the stations try to resend their frames after the timeout the frames will collide again Pure Aloha dictates that when the timeout period passes each user waits a

random amount of time before resending its frame the randomness will help to avoid more collisions this time is called as back off time TB the timeout period is equal to the maximum possible roundtrip propagation delay which is twice the amount of the time required to send a frame between the two most widely separated stations $2 \times TP$ let all the packets have the same length and each station requires one time unit for transmission which is TP consider any user to send packet a at the time T not if any other user B has generated a packet between time T not and T not plus TP the end of a packet B will collide with the beginning of packet a since in P Aloha packet a station does not listen to the channel

before transmitting it has no way of knowing that above frame was already underway this figure shows what vulnerable periods during which packets can collide similarly if another user wants to transmit between $t + TP$ and $t + 2 \times TP$ that is Packet C the beginning of packet c will collide with the end of packet a thus if two packets overlap by even the smallest amount of the vulnerable period both packets will be corrupted and need to be retransmitted let's calculate the throughput of Pure Aloha channel the throughput S of Pure Aloha is defined as an average successful traffic transmitted between the stations per unit time the unit of time is slot time which is the time required to transmit a frame assuming all frames or packets are

of same size since only one packet per slot can be transmitted the maximum value of S is 1 when collisions happen some of the packets are lost and part of available Channel time is wasted the resulting value of S is less than one to find the throughput of transmission Channel suppose the probability PK that K packets are generated during a given slot time follows a poisson distribution with the mean G per packet time is given by this equation the throughput S is then just the offered load G times the probability of a transmission being successful s is equal to $G \times P$ where p is equal to probability that a packet does not suffer a collision the probability of no other traffic being initiated during the entire vulnerable period is thus given

by $P = e^{-2G}$ therefore throughput s is equal to $G \times e^{-2G}$ the maximum throughput occurs at when G is equal to 0.5 therefore for Pure Aloha the best Channel utilization that can be achieved is around 18% no synchronization is required for Aloha each station can transmit a packet whenever it has one the disadvantage of this scheme is that it is inefficient because the maximum Channel utilization is only 18% of the total available capacity next is slotted Aloha in slotted Aloha the total Channel time is divided into discrete time slots and the stations can transmit only at specific instance of time these time slots are exactly equal to the packet transmission time all stations are then

synchronized to these time slots so that whenever a station has a packet it must synchronize exactly with the next possible time slot the benefit of this scheme is that the wasted time due to collisions can be reduced to one packet time or we can say that vulnerable period is reduced to half transmission attempts for four Network users and random retransmission delays for colliding packets in slotted Aloha are shown in this figure there are few assumptions in case of slotted Aloha all the frames are of the same size time is divided into equal size slots a slot equals the time to transmit one frame nodes start to transmit frames only at the beginning of a slot nodes are synchronized if two or more nodes

transmit in a slot all nodes detect Collision before the slot ends let's discuss the throughput of slotted Aloha channel in slotted Aloha the packets arrive in a synchronized fashion the probability of single transmission during a slot time is p not is equal to e^{-G} therefore throughput S is equal to G multiplied with e^{-G} the maximum throughput occurs at G equal to 1 which is twice that of Pure Aloha this means that the best Channel utilization that can be achieved is around 36.8% next technique is Carrier sense multiple access (CSMA) that is CSMA Carrier sense multiple access is a probabilistic media Access Control protocol in which a station confirms the absence of other traffic before

transmitting a data on a shared transmission medium carrier sense means that a transmitting station tries to find out whether another transmission is in progress before starting a transmission that is it tries to identify the presence of a carrier signal from another station before starting to transmit if a station senses a carrier the station waits for the transmission in progress to finish before starting its own Transmission in other words CSMA scheme is based on the principle of sense before transmit or listen before talk multiple access means that more than one stations may send and receive on the shared transmission medium the vulnerable time for CSMA is the propagation time T_P the propagation

time is the time required for a packet to propagate from one end to the other end of the medium now look at this figure in which station A transmits a frame at time T_1 which reaches the rightmost station D at time $T_1 + T_P$ this figure shows the vulnerable time for CSMA when a station senses the medium to be idle it transmits the packet by using one of these approaches nonpersistent CSMA one persistent CSMA and P persistent CSMA nonpersistent CSMA is a non-aggressive transmission protocol when the transmitting station has a data to transmit it first senses the medium if the medium is Idle then it transmits the data immediately but if the medium is busy then it waits for a random period of time during which it does not

sense the transmission medium before repeating the whole process that is sensing the transmission medium for idle or busy again this scheme reduces the collisions and which results in higher medium throughput next is one persistent CSMA one persistent CSMA is an aggressive transmission algorithm when the transmitting station is ready to transmit data it senses the medium for idle or busy if the medium is Idle then it transmits the data immediately but if the medium is busy then the station senses the medium continuously until it becomes

idle then it transmits the message unconditionally that is with probability equals to 1 in case of any Collision the sender waits for a random period of time and attempts to transmit

again unconditionally that is with probability equals to 1 one persistent csma is used in CSM CD systems including it ethernet next is p persistent CSM this is an approach between non-persistent and one persistent csma access schemes when the transmitting station has a data to transmit it then senses the transmission medium for idle or busy if the medium is Idle then it transmits the data immediately but if the medium is busy then it senses the medium continuously until it becomes idle then transmits a data frame with probability P if the station does not transmit data frame that is the probability of this event is 1 minus P then it waits until the next available time slot P persistent csma is

used in CS smca systems including Wi-Fi and other packet Radio Systems next technique is Carrier sense multiple aess with Collision detection that is CSM CD carrier sense multiple access with Collision detection is a media access control method used mostly in lens which are using early ethernet technology it uses a carrier sensing technology in which a transmitting station detects other signals in the medium while transmitting a frame and stops transmitting that frame transmits a jamming signal in case of collisions and then waits for a random period of time time before trying to resend the frame csms CD is a modification of pure carrier sense multiple access that is CSM scheme CSM CD is used to improve csma performance by terminating

transmission as soon as Collision is detected the base protocol is that a station with a message to send must monitor the channel to see if any other station is sending if another station is sending the second station must wait or 23 minutes, 1 second defer until the sending station has finished then it may send its message if no station was sending at the time that it first listened the station may send its message immediately the term carrier sensing indicates this listening before transmitting Behavior if two or more stations have messages to send at the same time and they are separated by a significant distance is on the bus each May begin transmitting at roughly the same time without being aware of the other station the signals from each station will superimpose on the channel

and is gobbled beyond the decoding ability of the receiving station this is termed as Collision let's discuss Collision detection procedure in csms the following procedure is used to resolve a detected Collision the procedure is complete when retransmission is initiated or the retransmission is aborted due to numerous collisions continue transmission with a jam signal instead of frame header data or CRC until minimum packet time is reached to ensure that all receivers detect the Collision increment re transmission counter was the maximum number of transmission attempts reached if so about transmission calculate and wait random back of period based on number of collisions re-enter main procedure at stage

one because of quick termination of transmission time and bandwidth is saved therefore CSM CD is more efficient than Aloha slotted Aloha and csma CSM CD Network Works work best on bus 25 minutes, 1 second multi-point topology with busy asynchronous transmission all stations are attached to One path and monitor the signal on the channel through transceiver attached to the cable csmd has totally decentralized control and is based on contention access csms CD supports both baseband and broadband system csmd offers four options in terms of bit rate signaling method and maximum electrical cable segment length these are 10 Bas 5 10 Bas 2 10 broad 36 and 1 base 5 the numeric field in the beginning indicates the bit rate in Mbps the middle term indicates the type

of signaling system that is baseband or Broadband the numeric field in the end indicates the electrical cable segment length in multi of 100 m Manchester signal code is used at the basement level of transmission in Broadband transmission different phase shift keying is used to convert the Manchester encoded signal into analog form so friends this was all about Mac Subler and Mac protocols hope all the concepts are clear to you hope to see you in the next lecture till then goodbye thank you